Brett Cavanaugh, Anne Jumper, & Ceci LaBarge
Prof. Gary Leisk
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Project 2 - Phase II

Introduction

Electric ratchet wrenches are compact electromechanical devices designed to deliver controlled torque through a high-reduction drivetrain while withstanding repeated cyclic loading and impact during fastening operations. Unlike manual ratchets, electric ratchets introduce additional design challenges due to the presence of a motor, gear reduction stages, and sustained torque output under continuous loading. As a result, the internal mechanical components must be carefully designed to balance strength, stiffness, and durability.

The objective of Phase II was to reverse-engineer and analyze the internal components of a commercial electric ratchet wrench to understand how mechanical loads are generated, transmitted, and absorbed within the device. The team fully disassembled the tool, created CAD models of the internal components within their subsystems in SolidWorks, identified the primary torque and reaction load paths, and conducted both analytical and numerical analyses of the system's key elements.

Based on their critical roles in torque transmission, structural support, and durability, the following machine elements were selected for further analysis: the bearing and outer casing assembly within the ratcheting head, the planetary gearbox, and the ratchet-and-pawl mechanism. These components enable the tool to transmit torque without slip, maintain shaft alignment under load, and withstand high-cycle fatigue and impact.

Each team member employed appropriate engineering tools from Engineering Design II and other courses, including free-body diagrams (FBDs), gear system analysis, fatigue and life estimation methods, finite element analysis (FEA), and motion analysis. The analyses were used to quantify stresses, deformations, contact pressures, safety factors, and the tool's predicted life.

Analysis - Bearings - Brett

Description

The component analyzed is the outer steel bearing casing (see Fig. 1) at the head of the electric ratchet wrench, along with the bearings housed within it. This is where the structural element is responsible for reacting to loads transmitted through the shaft and bearings from the drive rod. The tube houses two VXB HK122212 needle roller bearings (see Fig. 2) that support the driven shaft and react all radial loads generated by the ratcheting mechanism and gear train.

The casing houses two rolling-element bearings that support the shaft and maintain alignment during operation. Loads applied to the shaft are transferred through the bearing inner races, across the rolling elements, and into the outer races, which are adhesively bonded to the casing rather than mechanically retained. This load path makes the bearing seats and surrounding casing geometry critical regions for stress and potential fatigue failure.



Figure 1. Bearing Casing



Figure 2. VXB HK122212 Shell-Type Needle Bearing

Methods, Tools, and Techniques - Free Body Diagram

A combination of free-body diagram (FBD) analysis and finite element analysis (FEA) was used to evaluate the structural performance of the casing and bearing system. A simplified free-body diagram (see Fig. 3) was constructed in the primary plane of loading. The shaft was modeled as a beam supported by two bearings with a center-to-center spacing of 85 mm, measured between the outer faces of the bearings. Torque is applied to the shaft by the rotating drive rod through the ratchet mechanism located near the end of the casing. Because the drive rod engages the ratchet close to the outboard end of the housing, the resulting load is applied nearer to the ratchet rather than at the midpoint between the two bearings. Because of this, the bearing reactions are unequal and are obtained from the static equilibrium of a simplified shaft free-body diagram, which is then applied to the casing as equal and opposite loads at the bearing bores. Key assumptions for the FBD include:

- Predominantly radial loading with negligible axial load
- Quasi-static, repeated loading
- Identical HK122212 bearings with unequal load sharing
- Linear elastic material behavior
- Negligible friction, wear, and misalignment

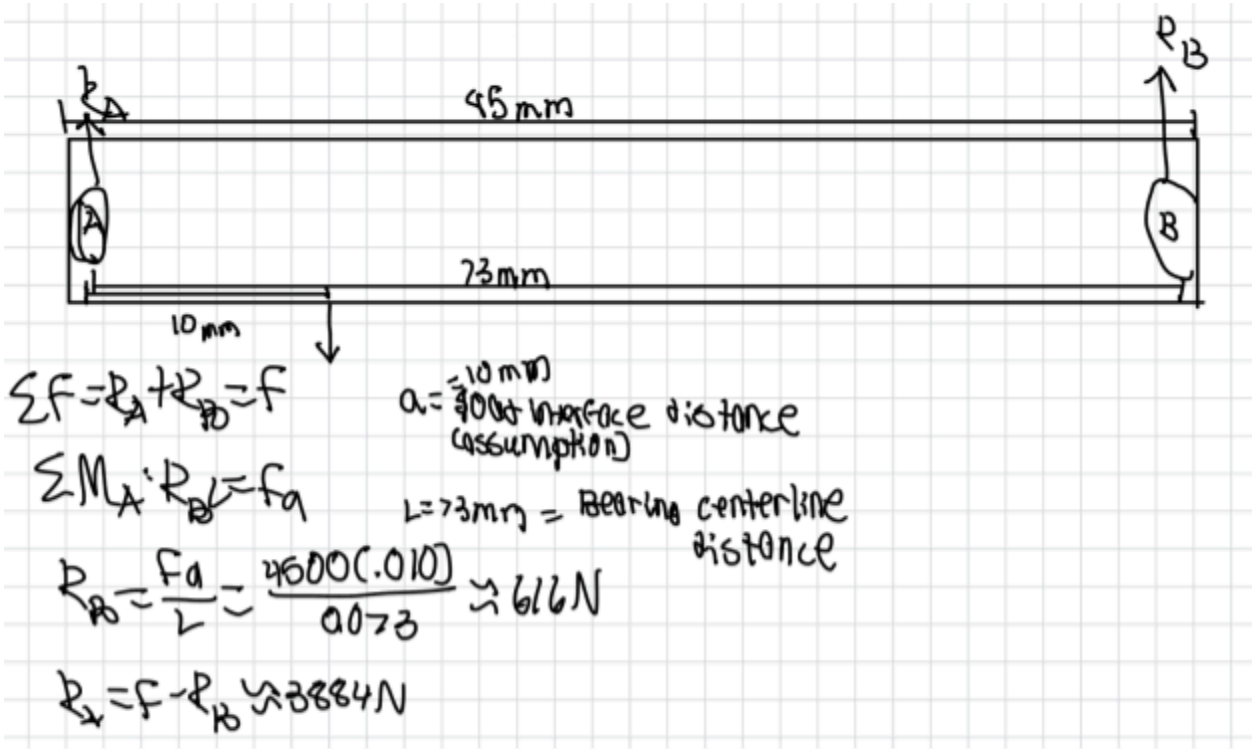


Figure 3. Free Body Diagram of Casing Assembly

Methods, Tools, and Techniques - Finite Element Analysis

For the finite element analysis, SolidWorks was utilized, and a linear static model was developed. The casing and bearing geometries (see Fig. 4), although simple, yield accurate results. Bearing supports were represented using bearing constraints applied to the cylindrical bearing bores, allowing rotation while restricting radial displacement. This approach captures the primary mechanical function of the bearings without explicitly modeling rolling contact. The casing was restrained at its mounting faces using fixed boundary conditions to represent attachment to the surrounding structure. These constraints prevent rigid-body motion while allowing internal stresses to develop in response to the applied bearing loads. Loads were used through the bearing constraints, corresponding to the reaction forces shown in the shaft's free-body diagram. This ensures consistency between the analytical load path and the numerical model. Bonded contact was assumed between the bearing outer races and the casing, representing a press-fit condition and preventing relative motion at the bearing-casing interface. AISI 1045 (carbon steel) was used as the material for the casing [2]. AISI 52100 (bearing steel) was used as the bearing material [1]. A mesh-convergence approach was employed, and Von Mises stress (see Fig. 5) and resultant displacement (see Fig. 6) were selected as the primary variables for evaluation. Key assumptions for the FEA include:

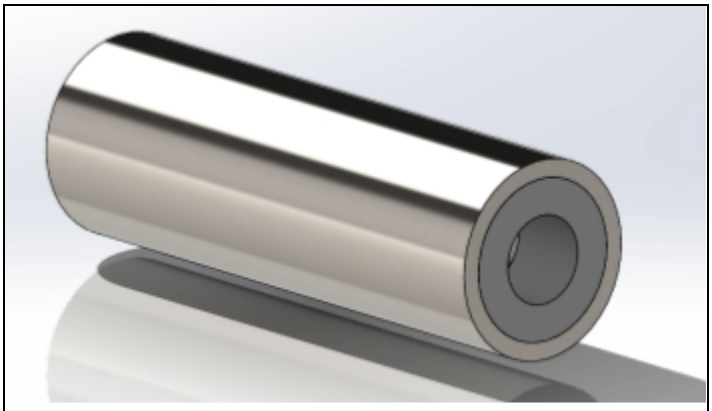


Figure 4. CAD Model of Casing Assembly

- Linear-elastic material behavior is assumed, as all predicted stresses remain well below the yield stress and no plastic deformation is expected.
- Static loading is assumed, with dynamic, inertial, and impact effects neglected. Repeated loading is addressed separately through fatigue considerations.
- Ideal bearing behavior is assumed. Bearings are modeled using bearing constraints that transmit radial load without friction, clearance, or misalignment.
- Perfect bonding is assumed between the bearing outer races and the casing, representing a press-fit installation.
- Symmetric load sharing between the two bearings is assumed, given the geometry and loading configuration.
- Manufacturing tolerances and surface roughness are neglected.
- Thermal effects are neglected, and material properties are assumed constant.

Given:
Casing Length: 85 mm = 0.085 m
Casing Outer Diameter: 27 mm = 0.027 m
Applied torque: T = 54 N.m
Bearing Axial Thickness: 22 mm
Effective radius: r = 12 mm = 0.012 m
Bearing spacing: L = 73 mm = 0.073 m
Overhang (assumption): a = 10 mm = 0.010 m

$$T = Fr \Rightarrow F = \frac{T}{r} \Rightarrow F = \frac{54}{0.012} = 4500\ N$$

$$\Sigma M_A = 0: R_B L = Fa$$

$$R_B = \frac{Fa}{L} = \frac{4500(0.010)}{0.073} = 616\ \text{N}$$

$$R_A = F - R_B = 4500 - 616 = 3884\ N$$

Results

The finite element analysis predicts a maximum von Mises stress (see Fig. 5) of approximately 19.6 MPa. This maximum stress occurs locally near the bearing bores and adjacent fillet transitions, which is consistent with the expected load transfer from the bearings into the casing. Stress levels elsewhere in the casing are significantly lower, indicating that the component is not globally stressed and the load transfer is localized. All predicted stresses remain within the elastic regime, and no yielding is observed.

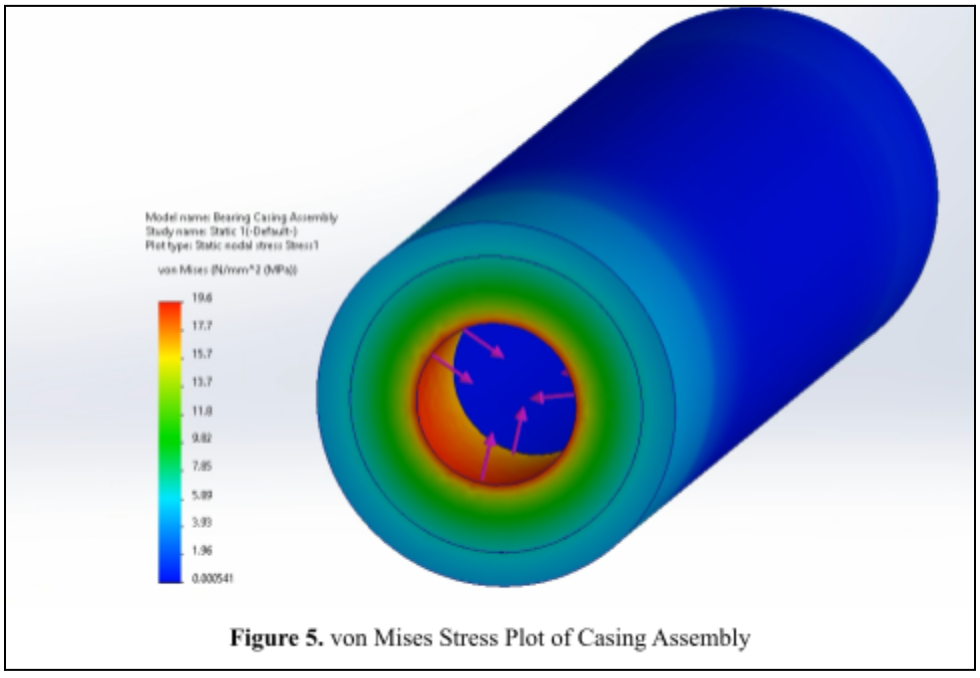


Figure 5. von Mises Stress Plot of Casing Assembly

The displacement plot (see Fig. 6) indicates a maximum resultant displacement of approximately 4.9 x 10^{-4} mm, occurring locally at the bearing bore where the radial load is

applied. Displacement decreases smoothly toward the fixed mountain faces, indicating stable load transfer and a stiff overall structure. The small magnitude of deformation confirms the validity of the linear-elastic, small-displacement assumptions used in the analysis. It suggests that bearing alignment is not significantly affected under the applied loading.

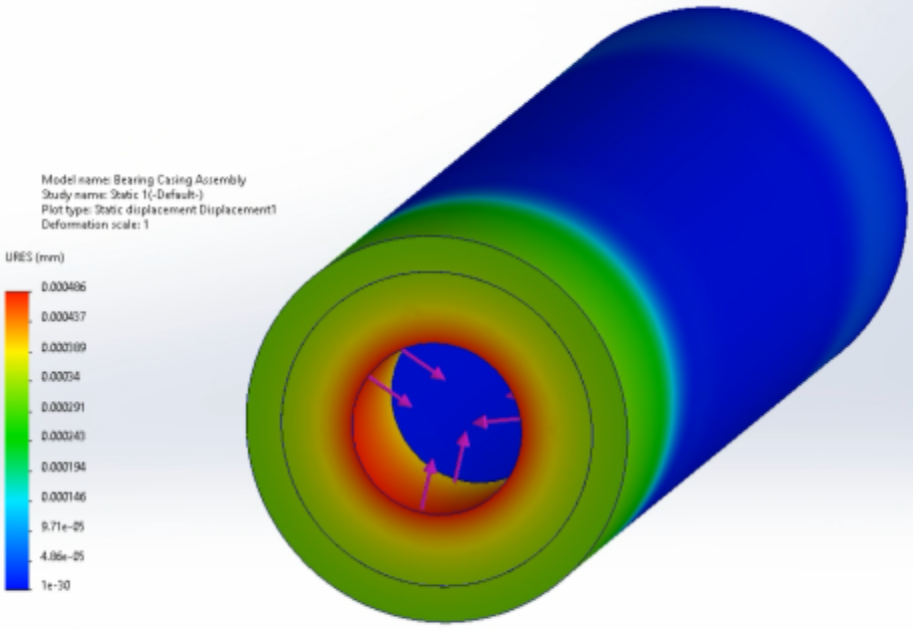


Figure 6. Resultant Displacement Plot of Casing Assembly

Although a detailed fatigue life calculation was not performed due to the absence of bearing manufacturer data and duty-cycle information, fatigue considerations were evaluated qualitatively. The maximum von Mises stress in the casing is approximately 19.6 MPa, which is well below the typical fatigue endurance limits for common structural materials at a design life of 10^6 cycles. Displacements are also minimal, indicating minimal risk of misalignment-induced fatigue failure. Bearing loads are shared between two supports and are not excessive, suggesting that bearing fatigue is unlikely to govern the design. Therefore, the casing and bearing system is expected to have adequate fatigue life for the intended application.

Discussion

The results indicate that the casing and bearing system is structurally adequate for the assumed loading conditions and design life. The maximum stress is low relative to typical material strength and fatigue limits, and stress concentrations are confined to small regions near the bearing seats. This suggests that the current design is conservative with respect to strength and fatigue.

The analysis also confirms that the casing, rather than the shaft, is the appropriate component to evaluate for structural adequacy, as it bears the primary reaction forces from the

bearings. The qualitative bearing life assessment further supports the conclusion that fatigue is unlikely to be a limiting factor.

Several limitations should be noted. Loading conditions and load locations were estimated, and ideal load sharing between bearings was assumed. Effects such as misalignment, impact loading, wear, lubrication degradation, and manufacturing tolerances were not included. Despite these limitations, the analysis is sufficient to demonstrate that the design meets structural and fatigue requirements within the scope of this project.

Analysis - Planetary Gearbox

Description:

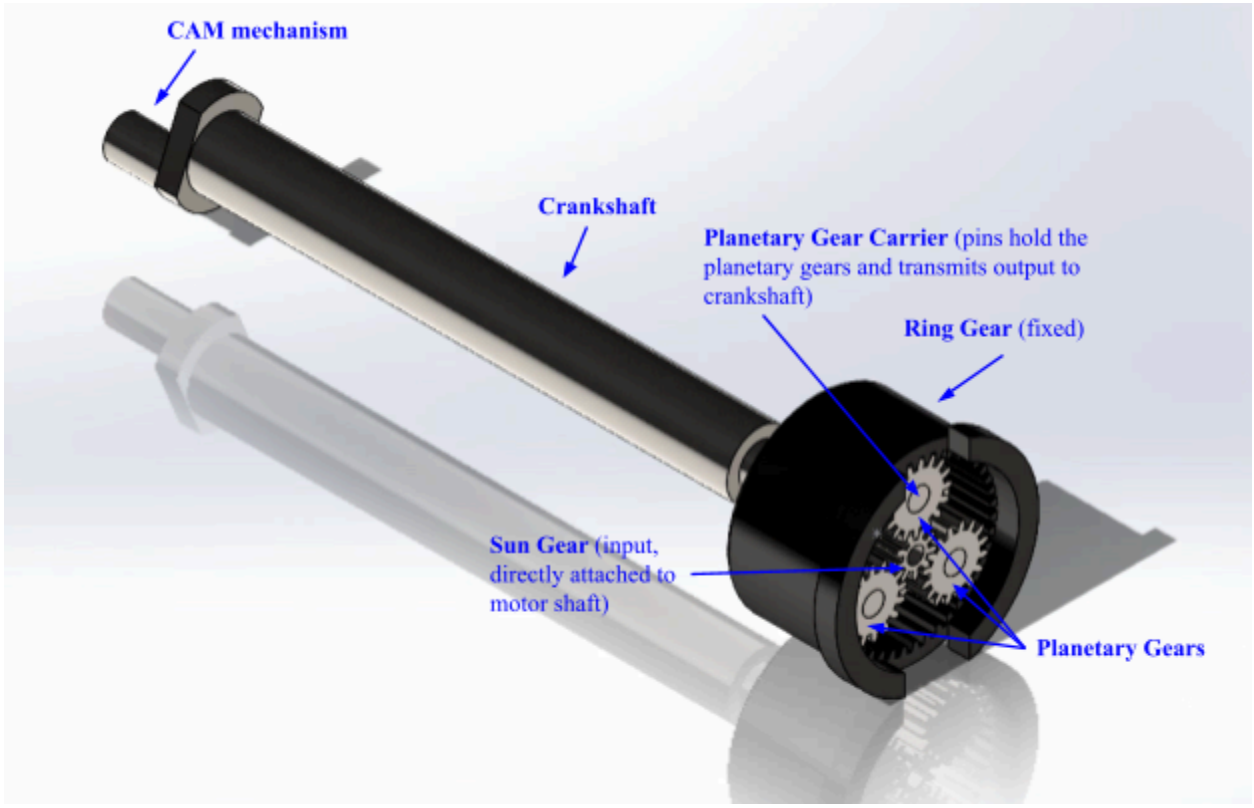


Figure 7. Diagram of planetary gearbox attached to crankshaft (drive rod). Not pictured is the motor which attaches directly to the sun gear.

The second element being analyzed is the planetary gearbox, which decreases speed and increases torque outputted by the motor. This torque amplification is essential to the functioning of the ratchet wrench since a significant amount of torque is required to tighten and loosen bolts.

However, it is important to note that the pawl/yoke mechanism of the ratchet system serves as an additional mode of torque multiplication, though not in the sense of a traditional gear ratio [3]. Rather, kinematic advantage is created by converting the continuous rotation of the crankshaft (drive rod) into oscillatory motion of the yoke which swings significantly but only advances the ratchet by a small angle. Since angular work can be calculated by multiplying torque by angular displacement, a smaller angular displacement requires a larger torque to conserve energy. Thus applying torque over a large internal angular motion and delivering it over a smaller output rotation results in a torque amplification and speed reduction to the 200 rpm maximum [4].

So, the planetary gearbox only does a portion of the torque amplification needed for the ratchet wrench to function as desired, but is still a significant component in achieving the 55 N.m maximum torque for screwing [4].

(a) (b) (c)

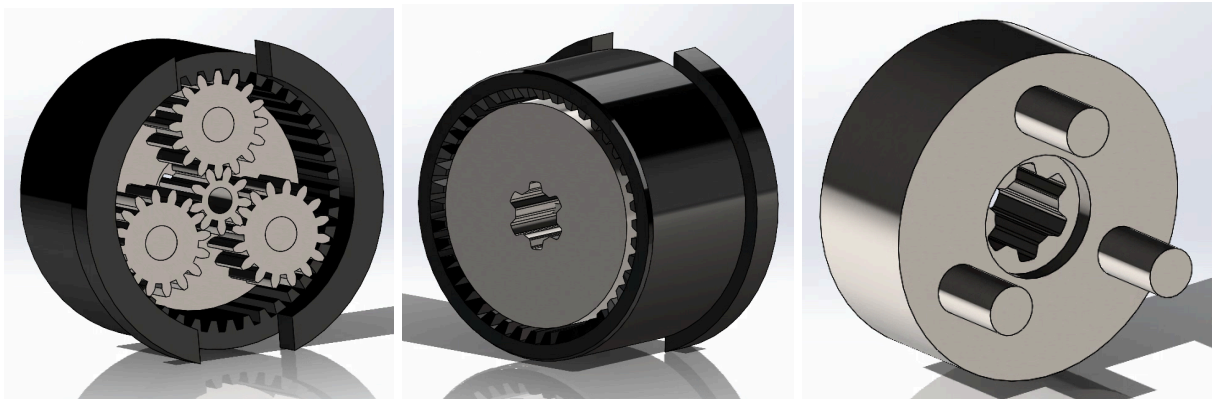


Figure 8. Planetary gearbox (a) front view of showing sun and planet gears and fixed ring gear, (b) back view showing back of carrier and slot for crankshaft (drive rod), and (c) planet gear carrier that transmits output of gearbox to crankshaft (drive rod).

This planetary gearbox consists of a single stage with input coming from the sun gear, a pinion with 9 teeth that is directly attached to the motor shaft to transmit the motor output and torque to the other elements of the gearbox. The sun gear meshes with 3 planet gears each with 16 teeth and evenly spaced around the sun. These planet gears rotate around axles that are attached to a single carrier (sometimes known as an arm). The carrier rotates, serving as the output of the gearbox which the crankshaft (drive rod) is attached to. The planet gears mesh with a fixed outer ring gear with 42 teeth that fits snugly to the outer motor casing. The ring gear allows the carrier to rotate with the planet gears.

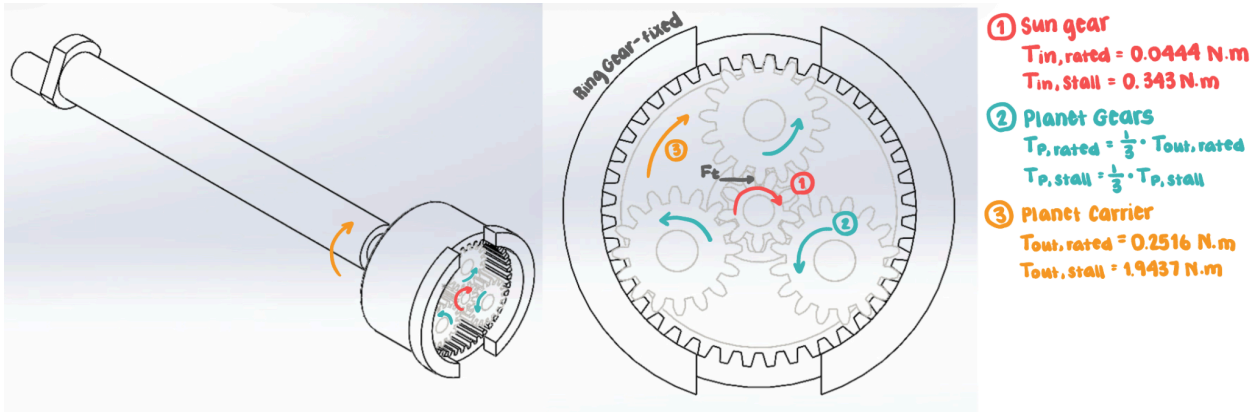


Figure 9. Free body diagrams of the planetary gearbox, depicting the results of the torque analysis.

Analytical Analysis:

The analytical analysis begins with identifying the gearbox's geometry, which consists of a 9-tooth sun gear, three 16-tooth planet gears, and a 42-tooth fixed ring gear. The tooth count does not satisfy the standard $N_R=N_S+2N_p$ relationship for planetary gear systems, which may indicate that the manufacturer used profile shifting to achieve the desired meshing and center distance (which is common in creating compact commercial gearboxes). For a single stage planetary gear system, like that of the ratchet wrench, the RPM of the ring gear, carrier, and sun gear are related by the following equation:

$$n_R \cdot N_R = n_C \cdot \left(N_R + N_S\right) - n_S \cdot N_S$$

Here, n_R is the speed of the ring gear, N_R is the number of teeth in the ring gear, n_C is the speed of the carrier, n_S is the speed of the sun gear, and N_S is the number of teeth in the sun gear. In this case, with input coming from the sun gear and a fixed ring gear ($n_R = 0$), the speed (and torque) ratio is given by:

$$G = \frac{n_S}{n_C} = \frac{N_R+N_S}{N_S} = \frac{42+9}{9} = \frac{51}{9} \approx 5.667$$

Voltage		No Load		Rated Load					Stall	
Operating Range	Rated	Speed	Current	**Speed**	Current	**Torque**	Output	Efficiency	**Torque**	Current
VDC		RPM	A	**RPM**	A	**N.m**	W	%	**N.m**	A
6.0 - 14.0	12	18000	1.25	**15669**	8.4	**0.0444**	72.86	72.3	**0.343**	56.5

Table 1. Excerpt from data sheet for RS550 12V brushed DC motor with relevant metrics in boldface. [5]

Using the motor datasheet [5] excerpted in table 1, the input torque was taken as 0.0444 N.m at rated operation and 0.343 N.m at the stall. Using the torque ratio calculated above, the output torque from the gearbox exerted on the crankshaft (drive rod) is calculated as follows:

$$\text{Rated: } T_{out} = G \cdot T_{in} \cdot \eta = 5.667(0.0444\ N.m)(1) = 0.2516\ N.m$$

$$\text{Stall: } T_{out} = G \cdot T_{in} \cdot \eta = 5.667(0.343\ N.m)(1) = 1.9437\ N.m$$

Note that the gearbox efficiency is assumed to be 100% since the steel components are well-lubricated and because we aim to analyze gear performance in worst-case torque scenarios to ensure the design is adequate. Note that the efficiency given in table 1 is that of the motor itself while the efficiency used here is of the gearbox. Note that these torques are significantly less than the 55 N.m maximum the ratchet wrench boasts [4], exemplifying the significant role of the yoke and ratchet mechanism in further amplifying the torque.

In estimating internal loads in the planetary gearbox, unequal load sharing between the three planets is assumed, with a single planet carrying up to 50% of the total gearbox output torque. This assumes worst-case conditions when manufacturing tolerances or gear misalignment prevent the three planet gears from equally carrying torque.

Additionally, the forces between the sun and planet gears are analyzed most closely since these gears are most concerning for failure due to their small sizes (few teeth and thus higher tooth stresses and high torque due to small radii). The tangential tooth force at the planet–sun mesh is calculated using the following equation:

$$F_t = \frac{2T_{in}}{d} = \frac{2(50\% \cdot 1.9437\ N.m)}{0.0111\ m} = 175.045\ N$$

The gearbox output torque is used since the planetary stage as a whole must transmit this force internally. Also, since tangential force exerted by the planet gear on the sun gear is being calculated, the diametral pitch of the planet gear is used. The module was measured from the physical gear (0.694mm) to determine the diametral pitch using the following formula:

$$d = m \cdot N_p = (0.000694\ m)(16) = 0.011104\ m$$

This tangential force is then used as the applied load in the static finite element analysis of the planet gear tooth to evaluate tooth-root bending stresses and assess the structural adequacy of the gearbox.

FEA Stress Analysis:

A static FEA simulation was performed on a simplified model of the gearbox to see the sun gear's interaction with a single planet gear. The sun-planet interface was chosen since it experiences high tangential tooth forces as this force is inversely proportional to the pitch radius of the driving gear, which is extremely small for the sun gear. Additionally, the small size of these gears and their small surface areas lend themselves to high stresses, making them prone to failure.

To evaluate the worst-case condition, the output torque from the gearbox is used in the stall condition, since the motor produces its maximum torque (lowest speed) at stall. The gearbox output torque is used instead of the motor torque because the gear teeth must transmit the torque after it has been multiplied by the gear reduction, making the output torque the governing load for tooth stress. Additionally, unequal load sharing between the planets is assumed in analyzing a worst-case condition, with an assumed 50% of the total gearbox torque being exerted on a single planet gear.

(a) (b)

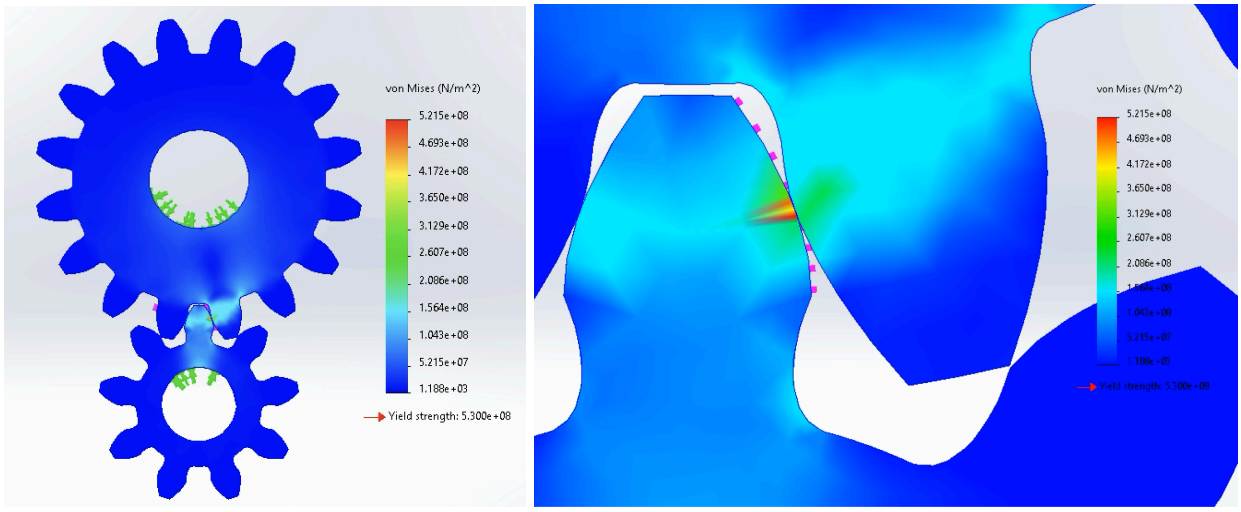


Figure 10. von Mises stress (N/m^2) colored contour plot for the sun-planet gear system (a) as a whole and (b) at the interface.

With these assumptions the 175.045 N tangential tooth force is applied to the top tooth of the smaller sun gear on the left side. The load is applied uniformly across a small area around the line of contact between the gears with direction normal to the tooth surface. The bore of the planet gear is fixed to all translation and rotation while that of the sun gear is fixed to translation, but is free to rotate about its central axis. Additionally, contact component interaction is used so that the gears are constrained from penetrating through one another. The mesh applied to the system is moderately coarse with refinement on the right side of the sun gear, since localized high stress is at this interface and refinement improves accuracy and reduces artifacts from contact at a single point. Finally, the gears are assumed to be made of AISI-1045 cold-drawn steel with a yield strength of 5.3e+08 N/m^2, since it is commonly used to make parts like gears due to its high tensile strength and good machinability [6].

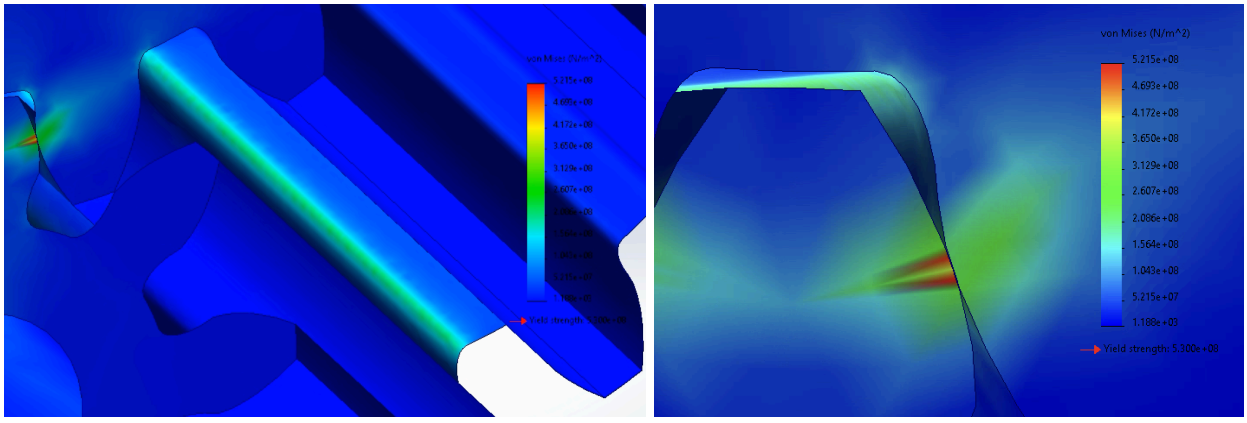


Figure 11. von Mises stress (N/m^2) colored contour plot at the planet-sun gear interface, focusing on high-stress regions at the tooth roots of the planet gear.

The static FEA simulation indicated that highest von Mises stresses occurred at the tooth root fillets, where stress is probed to be as high as 2.921e+08 N/m^2. These results can be seen in figure 11. This location of high stress is consistent with typical gear tooth bending failure theory. Although the sun gear is the smaller input member and was the location where the load was applied, the planet gear exhibited higher overall tooth-root stresses due primarily to its smaller face width (7 mm compared to 10 mm for the sun gear), resulting in higher stress for the same transmitted force. Localized regions of very high stress are observed at the gear tooth contact region (see figure 10(b)), and are near the material yield strength. However, these peaks can likely be attributed to simplification of gear contact assumptions and simulation artifacts, since refining the mesh in this region reduced stress significantly compared to my initial model without mesh control. Thus, the tooth-root bending stress was instead taken as the governing design metric.

Results:

Solutions obtained analytically indicate that the planetary gearbox effectively transmits torque by reducing the motor output and its high speed and low torque to a lower speed and higher torque output at the carrier which is attached to the crankshaft (drive rod). Namely, torque is magnified by a factor of 5.667 as indicated by the torque ratio calculation. Under the rated/stall loading conditions of the motor, its 0.0444 N.m/0.343 N.m output torque is converted to a 0.2516 N.m/1.9437 N.m torque by the gearbox.

The results of the static FEA stress analysis predicted a maximum von Mises stress of 2.921e+08 N/m^2 at the tooth root fillet of the planet gear, the critical region for gear tooth bending failure. Using the material yield strength for AISI-1045 cold-drawn steel of 5.3e+08 N/m^2, the corresponding factor of safety (FoS) for steel, a ductile material, is calculated to be about 1.8:

$$FoS = \frac{Yield\ Stress}{Working\ or\ Design\ Stress} = \frac{5.3\times10^8\ N/m^2}{2.921\times10^8\ N/m^2} = 1.8144$$

This factor of safety is calculated under the assumed worst-case loading condition and still maintains a value greater than 1, indicating a safe design since the capacity is greater than the demand. Under operating conditions, we can expect this factor of safety to be much higher since the calculated stall torque (1.9437 N.m) of the carrier is much higher than the rated torque (0.2516 N.m), thus producing higher teeth stresses.

Discussion:

The FEA results indicate the adequacy of the design of the gearbox components, since the peak stresses at the sun-planet tooth interface remain below the material yield strength even under worst case conditions (i.e. stall torque conditions and unequal load sharing between planets). However, localized regions of high stress at the contact regions, which are likely due to simulation artifacts, may also indicate the importance of surface treatment for these machine elements.

However, the analysis performed was limited in several ways. Only a small portion of the gearbox (the sun and a single planet gear) were modeled using FEA, which omits interactions between the planets, the carrier and planet gear pins, and the ring gear. These elements could affect the load distribution or could also be regions of high stress concentration prone to failure.

Additionally, practical constraints such as precise measurement of physical gears and slight modifications to the gear module to ensure proper meshing in the FEA simulation may have altered geometry, namely the gear diameter, and thus stress predictions. While the simulation and analytical results indicate an adequate design, real-world stresses and fatigue effects could indicate otherwise. Further analysis of the gearbox as a whole along with real-world experiments would assess the component performance more comprehensively.

In future investigations, it would be interesting to do a broader analysis of torque transmission within the device, including additional torque multiplication from the ratchet mechanism. Elements of the device where torque is highest would be interesting to perform a static stress analysis on to observe other regions of likely torque-related failure in the wrench.

Analysis - Ratchet - Ceci

Description

The third component analyzed was the ratchet system, which is the primary element that transmits torque from the electric motor to the bolt. Inside the power ratchet wrench housing is a standard ⅜” square drive ratchet head (Figure 12, Part 1). The ratchet head is mounted on a cylindrical pivot pin (Figure 12, Part 3) that is concentric with the head but offset from the drive rod axis. The offset pin allows the head to rotate about the y-axis while keeping the front face parallel to the front plane, pivoting around the cylindrical pivot pin, as seen in figure 13.

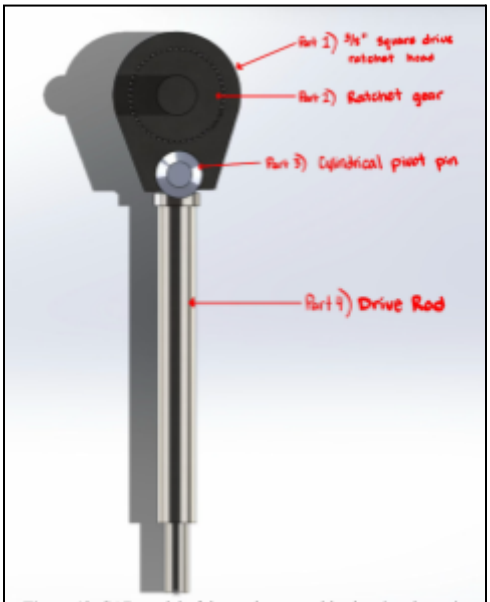


Figure 12. CAD model of the ratchet assembly showing the main components: (1) 3/8" square-drive ratchet head, (2) ratchet gear, (3) cylindrical pivot pin, and (4) drive rod.

When the 12 V DC motor spins the driving rod (Figure 12, Part 4) in rotary motion through the planetary gearbox, the offset connection causes the ratchet head to rock back and forth relative to the rod. As the ratchet head rocks, the ratchet gear (Figure 12, Part 2), rotates relative to the housing and its surrounding teeth. In the actual tool, a spring-loaded pawl (not represented in the CAD) engages these teeth and, depending on its orientation, allows the gear to drive in either the tightening or loosening direction while free wheeling in the opposite direction.

For this analysis, we focused on the rotary motion of the ratchet gear and the torque transmitted through the motor/rod system. To make the problem tractable, the detailed pawl mechanism was replaced with an equivalent simplified model that produces the same effective torque to the ratchet head.

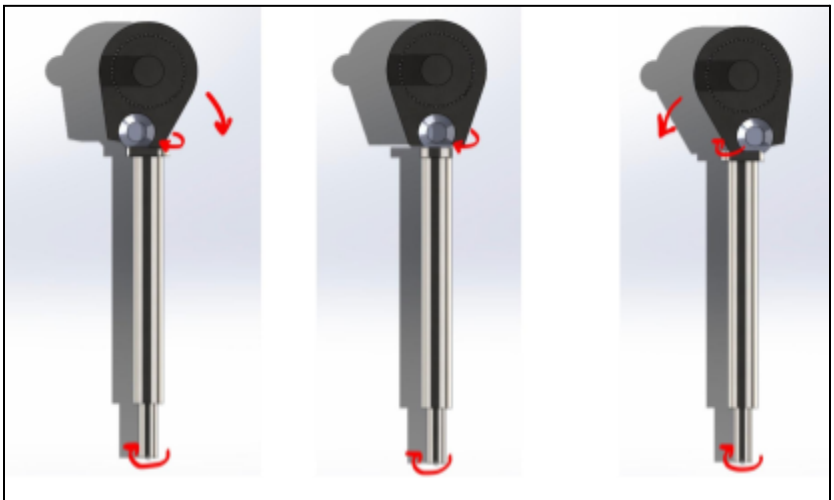


Figure 13. Sequence of ratchet head positions during rotation of the drive rod, showing how the head tilts relative to the rod as torque is applied.

Free Body Diagrams

Figure 14 shows a free-body diagram of the ratchet head for case 1 (bolt engagement). In this case, the ratchet tightens or loosens an engaged bolt, so the head experiences an applied torque from the bolt, an opposing motor torque on the drive rod, and reaction forces at the pivot. Figure 15 shows a free-body diagram of the ratchet head for case 2 (no bolt engagement). Here, the ratchet is powered but not engaging a bolt, so no torque is transmitted through the inner gear, and no tooth-contact forces are present. Compared with case 1, the only remaining external loads are the motor torque on the drive rod and the corresponding reaction forces at the pivot.

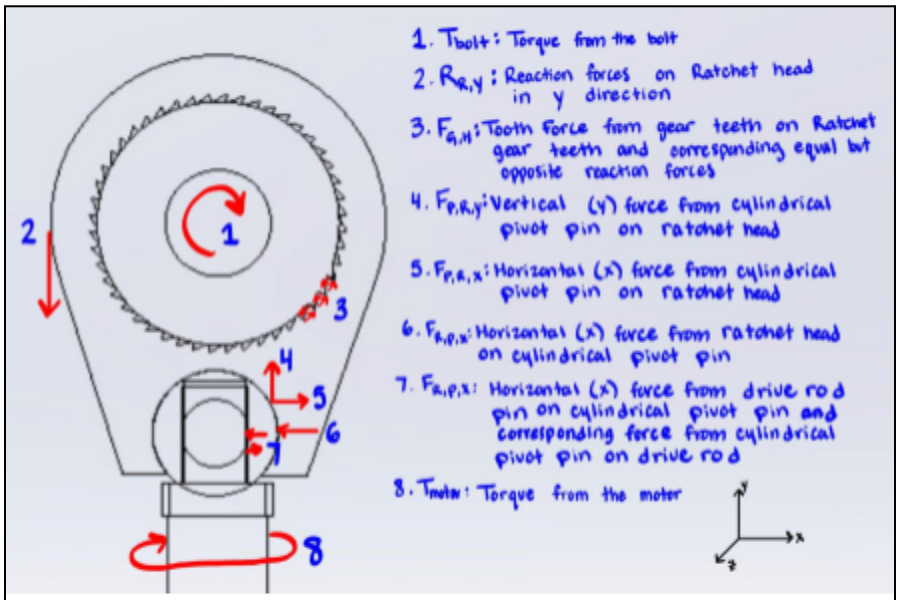


Figure 14. Free-body diagram of the ratchet head for **Case 1** (bolt engagement): maximum torque condition, with the head about to tilt toward the left.

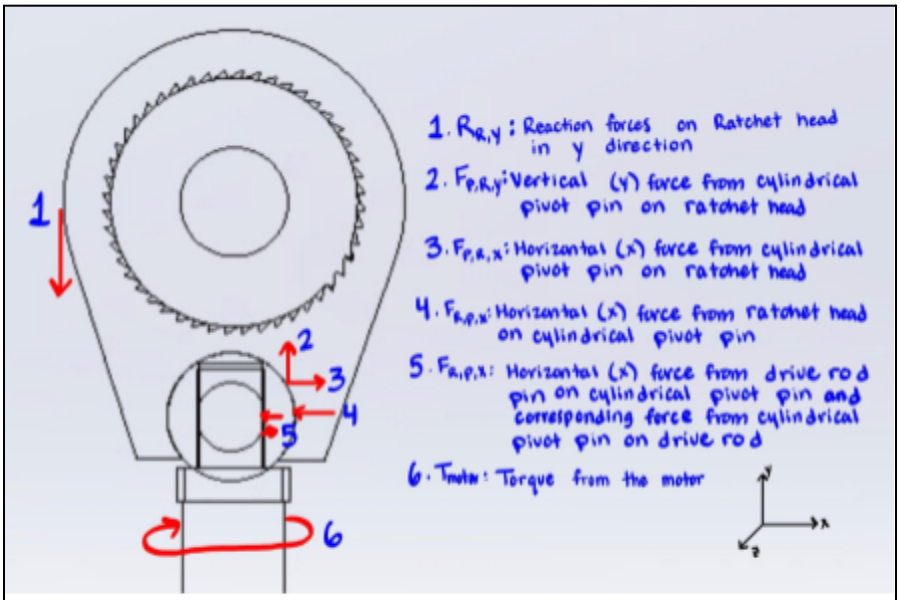


Figure 15. Free-body diagram of the ratchet head for **Case 2** (no bolt engagement): maximum torque condition, with the head about to tilt toward the left.

Motion Analysis

To perform the motion analysis, we first modeled the ratchet system in CAD. The geometry is representative of the real tool, but includes a few simplifications to make the analysis tractable. After creating each part, we assembled the model with appropriate mates and small clearances to avoid redundant constraints and to ensure realistic motion. Based on product information and reasonable assumptions, all components were modeled as cast stainless steel. We then set up two separate motion studies. In case 1 (no bolt engaged), the ratchet head was free, and only loading was provided by a 12 rpm rotary motor applied to the drive rod, with solid-body contact forces (with the material being greasy steel) defined between the interacting parts. The study length was trimmed to focus on the time interval around the peak motor torque. In case 2 (bolt engaged), we used the same setup but also applied a clockwise torque at the bolt attachment location to represent engagement with a fastener.

We ran two motion studies for case 1. In the first study, we applied the full rated bolt torque of 40 lbf*ft at the ratchet head (where the bolt would be attached). The rotary motion was set to 12 rpm. As shown in figure 16, the motor torque increases as the ratchet head swings into the loaded position, reaches a peak torque of about 921 N/mm (~0.68 lbf*ft) at ~0.65 s, and then decreases as the ratchet heads continue on its cycle. The configuration of the ratchet head at maximum torque is also shown in figure 16.

In the second study, we repeated the analysis with 20 lbf · ft applied at the head. As expected, the peak motor torque decreased to about 718 N/mm (~0.53 lbf*ft) at ~0.65 while the overall shape of the torque time graph curve remained similar. Again, the configuration of the ratchet head at maximum torque is also shown in figure 17. Finally, for case 2, we conducted a baseline study with no torque applied to the head (i.e., no bolt engaged). In this case, the peak motor torque was approximately 60 N · m (~0.04 lbf · ft) (shown in Figure 18). This small but non-zero value is due to inertia, contact, and friction in the mechanism. Overall, the results are consistent; the motor torque increases with applied bolt torque, peaks when the ratchet head passes through its maximum torque stage, and then decreases as the head continues to rotate.

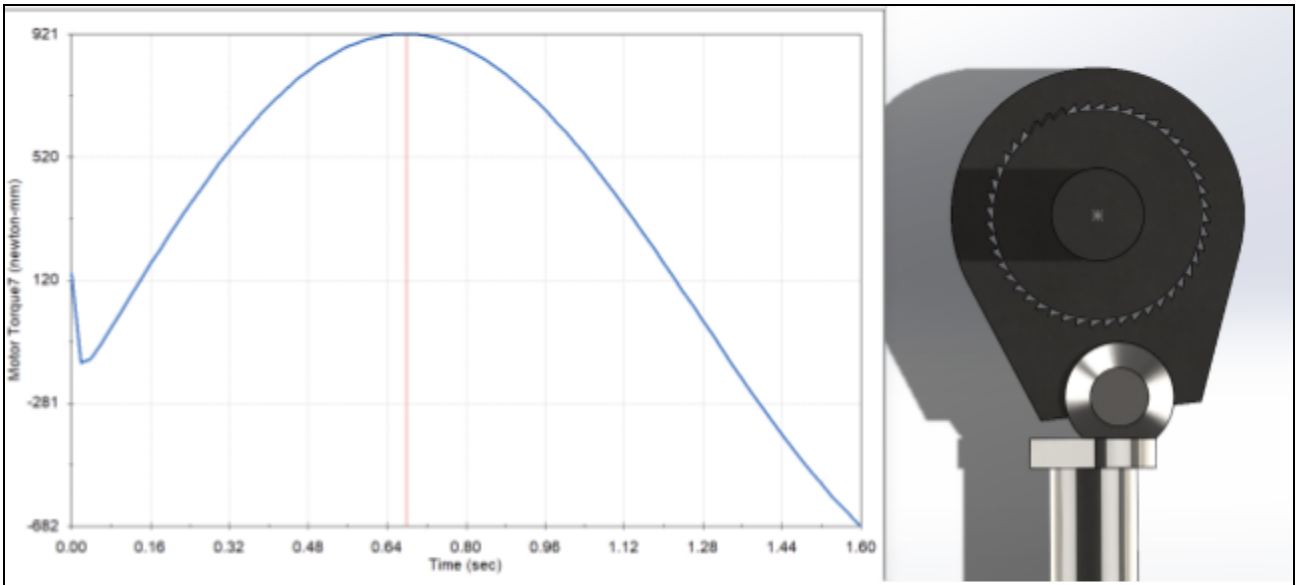


Figure 16. Motor Torque With 40 ft·lbf Torque Applied at the Head & Max Torque Position Shown

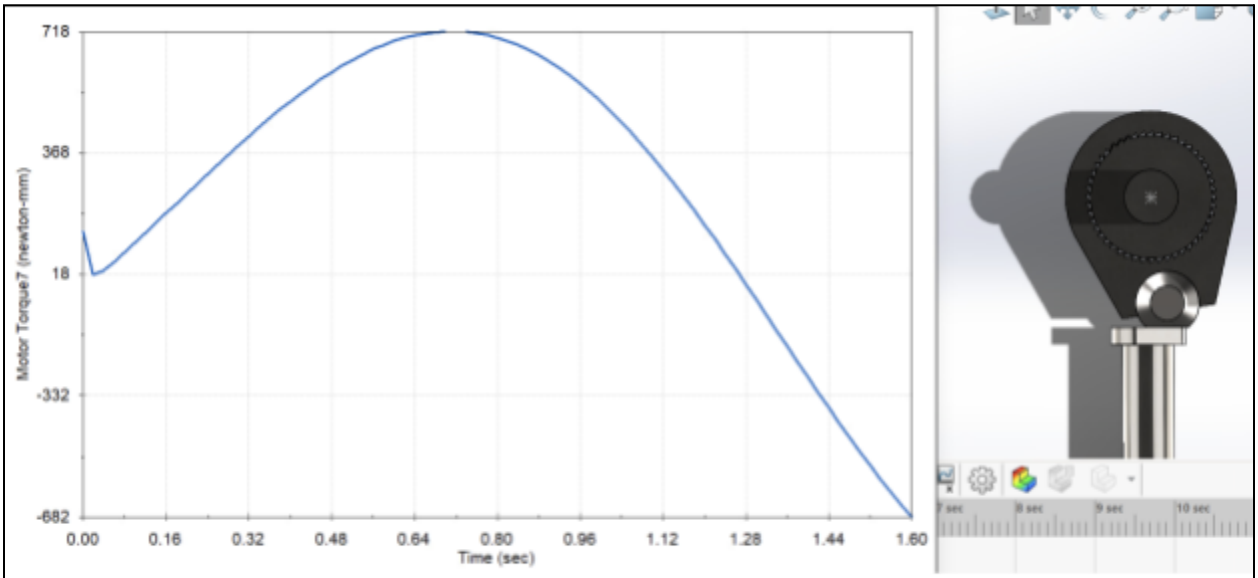


Figure 17. Motor Torque With 20 ft·lbf Torque Applied at the Head & Max Torque Position Shown

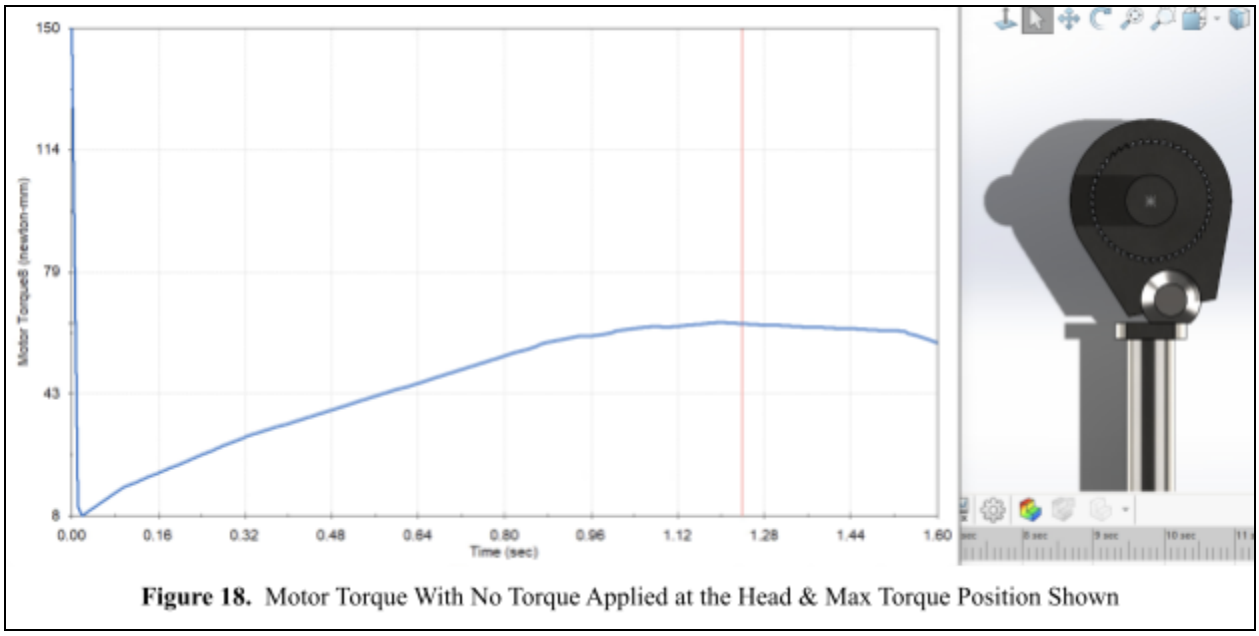


Figure 18. Motor Torque With No Torque Applied at the Head & Max Torque Position Shown

[Video of motion analysis no bolt engaged]

[Video of motion analysis Bolt Engaged (Max Torque Applied)]

FEA

We developed an FEA model of the ratchet system under a maximum torque of 40 lbf*ft. To do this, we defined the appropriate fixtures and contact interactions and applied local mesh controls in the high-stress regions (Figure 19). Using this torque, we first calculated the bending stress in the engaged gear tooth by hand to validate our SolidWorks simulation. From the hand calculations shown in figure 20, we found the maximum ending stress on the middle gear tooth to be $9.71*10^8$ N/m^2. This compares well with the SolidWorks FEA results. When probing the stress at the root of the same tooth in the SolidWorks model (Figure 21), we obtained a von Mises stress of $9.04*10^8$ N/m^2. This value is relatively close to the hand calculations (within 7%) therefore it is reasonable, given the geometric simplifications in the hand calculation. This close agreement provides confidence that our simulation accurately captures tooth stresses.

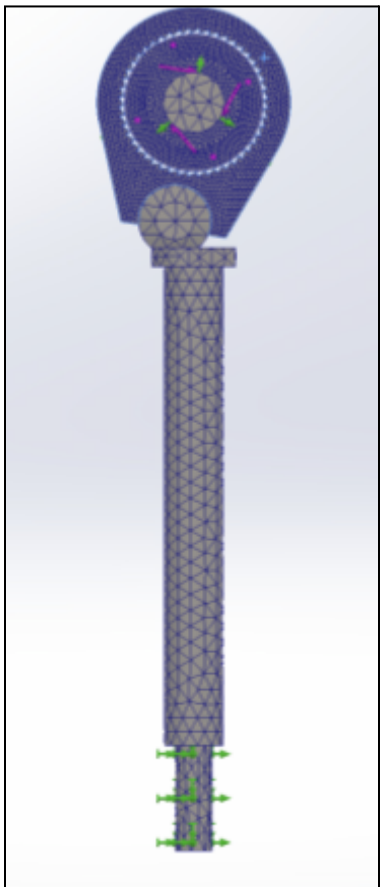


Figure 19. Mesh for FEA

Calculate stress on gear tooth

$F_t = \frac{T_{max}}{r_p} = \frac{54\,000\ N mm}{14.76\ mm} = 3658.53\ N$

$I = \frac{bt^3}{12} = \frac{(13.11)(1.3)^3}{12} = 2.4$

$\sigma_b = \frac{Mc}{I} = \frac{F_t h (t/2)}{I} = \frac{(3658.53)(0.98)(1.3/2)}{2.4} = 971\ N/mm^2$

$T_{max} = 40\ lbft \rightarrow 54,000\ N mm$

$h = 0.98\ mm$

$t = 1.3\ mm$

$b = 13.11\ mm$

$r_p = 15.25 - 0.49 = 14.76\ mm$

$\sigma_b = 971\ N/mm^2 = 9.71 \times 10^8\ N/m^2$

Figure 20. Calculations of Stress on Engaged Gear Teeth

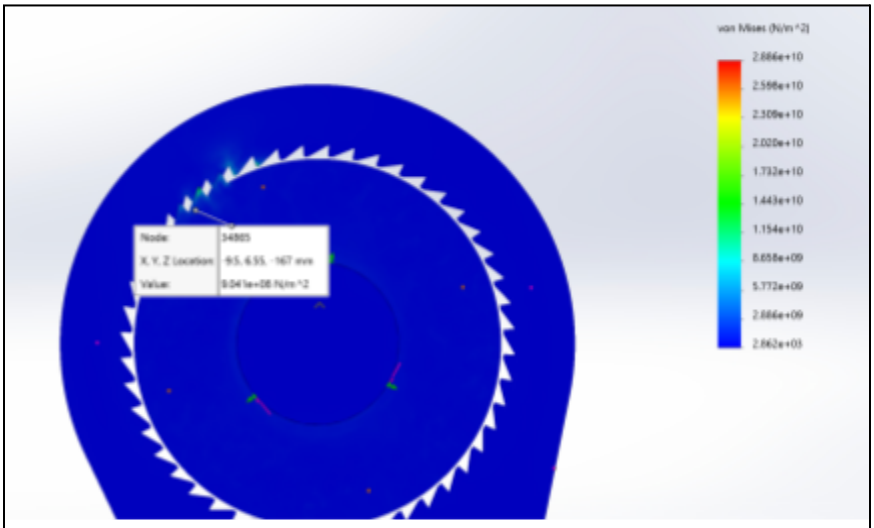


Figure 21. Probed Stress Value in Middle Tooth Found to be $9.041*10^8 N/m^2$

Conclusion

Reverse engineering and analysis of the electric ratchet wrench provided a clear picture of how torque is generated, amplified, and transmitted through the device, and thus where it is likely to fail. Each of us analyzed a different subsystem, but doing so revealed how tightly all the systems are connected and how changes in one part affect the others.

The casing and bearing system was analyzed using a combination of free-body diagram analysis and finite element modeling to evaluate load transfer, stress, and deformation under applied torque from the ratchet mechanism. Bearing reaction forces were determined from a simplified shaft free-body diagram, assuming overhung loading from the torque input applied near the end of the casing. These reactions were applied to an FEA model of the casing to evaluate structural response. The results show low stress (~19.6 MPa) and very low displacement (4.9×10^{-4} mm), indicating that the casing design is structurally sound for the assumed loading conditions, with fatigue and bearing life not expected to govern performance within the scope of this analysis.

The planetary gear system used to amplify torque from the motor was analyzed using both analytical analysis of torque transmission and FEA analysis of tooth stresses resulting from the amplified torque. First, the torque ratio was calculated using the tooth counts of the gears (5.667). Multiplying this value by the rated and stall torques produced by the motor yields the output torques from the gearbox in rated and stall loading conditions: 0.2516 and 1.9437 N.m (assuming 100% efficiency for worst-case torque). The stall (worst-case) output torque was used to find the tangential tooth force exerted by the planet gear on the sun gear (175.045 N) and this value was used in an FEA simulation of static stress to analyze gear teeth stresses. Stress was observed to be maximized at the tooth root fillet of the planetary gear ($2.921e+08$ N/m^2), giving a safety factor of 1.8 with the material properties of carbon steel. This value is less than 1, indicating an adequate design by manufacturers even in the worst case.

The ratchet-head motion analysis and FEA are integrated at the output. The motion analysis showed that when 40 lbf · ft is applied at the head, the motor requires ~0.7 lbf · ft to sustain motion, and this torque scales appropriately when the applied torque is halved or removed, consistent with the ratchet's mechanical advantage. The hand calculation of bending stress in the engaged gear tooth ($9.71*10^8$ N/m^2) agreed within about 7% of the FEA stress ($9.04*10^8$). This gives us confidence that our simplified models capture the main forces acting on the ratchet gear teeth.

Overall, the results suggest that the commercial design is structurally robust in its housing and bearings, while the most likely mechanical weak points under abusive use are the highly loaded gear teeth in the planetary set and the ratchet head, particularly at the roots and contact surfaces. The wrench appears well-designed for its intended torque rating, with reasonable safety margins, but it is clearly dependent on proper lubrication, manufacturing tolerances, and surface hardness to avoid gear and pawl wear over many cycles.

References

[1]Corner, L., 2013, “Aisi 52100 alloy steel (UNS G52986),” AZoM [Online]. Available: https://www.azom.com/article.aspx?ArticleID=6704. [Accessed: 15-Dec-2025].

[2]Corner, L., 2025, “Aisi 1045 Medium Carbon Steel,” AZoM [Online]. Available: https://www.azom.com/article.aspx?ArticleID=6130. [Accessed: 15-Dec-2025].

[3]Peace, Jonathan. “How Does a Power Ratchet Work?” *Ingersoll Rand*, 2 July 2025, https://powertools.ingersollrand.com/en-us/news/how-does-a-power-ratchet-work/. Accessed 15 December 2025.

[4]*Original Operating Instructions 3/8″ Extended Cordless Ratchet Wrench*. WorkPro.

[5]“Johnson Electric RS-550 Motor Replacement Motor 12V 21000RPM High Speed - 550 & 555 Size DC Motor.” *micro pmdc motor*, https://www.foneacc-motion.com/product/Johnson-Electric-RS-550-Motor-Replacement-Motor-12V-21000-RPM-High-Speed-550-555-Size-DC-Motor.html. Accessed 15 December 2025.

[6]McClements, Dean. “1045 Carbon Steel: Uses, Composition, Properties.” *Xometry*, 7 July 2023, https://www.xometry.com/resources/materials/1045-carbon-steel/#:~:text=It%20is%20suitable%20for%20manufacturing,common%20applications%20for%201045%20steel. Accessed 15 December 2025.